① $\hat{H}\hat{p}_x \phi(x) = \hat{H}\left(-i\hbar \phi'(x)\right)$

$$= \left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + \frac{1}{2}kx^2\right)\left(-i\hbar\phi'(x)\right)$$

$$= \frac{i\hbar^3}{2m}\phi'''(x) - \frac{i\hbar k x^2}{2}\phi'(x)$$

$$\hat{p}_x\hat{H}\phi(x) = \hat{p}_x\left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}+\frac{1}{2}kx^2\right)\phi(x)$$

$$= \left(-i\hbar\frac{\partial}{\partial x}\right)\left(-\frac{\hbar^2}{2m}\phi''(x)+\frac{1}{2}kx^2\phi(x)\right)$$

$$= \frac{+i\hbar^3}{2m}\left(\phi'''(x)\right) - \frac{i\hbar k x^2}{2m}\phi'(x)$$

$$- i\hbar kx\,\phi(x)$$

$$[\hat{H},\hat{p}_x]\phi(x) = \hat{H}\hat{p}_x\phi(x) - \hat{p}_x\hat{H}\phi(x)$$

$$= i\hbar kx\,\phi(x)$$

$$[\hat{H},\hat{p}_x] = i\hbar kx$$

Q.2 (i) $\frac{d}{dx} e^{ikx} = ikx \cdot e^{ikx}$

e^{ikx} is an eigen function of operator $\frac{d}{dx}$

(ii) $\frac{d}{dx} e^{\alpha x^2} = \underbrace{2\alpha x}_{\text{not a constant}} \cancel{x^2} \cdot e^{\alpha x^2}$

So, $e^{\alpha x^2}$ is not an eigen function

Q.3 $E_n = \frac{n^2 h^2}{8 m L^2}$

$E_2 - E_1 = \Delta E = \frac{3h^2}{8mL^2} = 18.06 \times 10^{-24} J$

$\Delta E = \frac{hc}{\lambda}$

$\Rightarrow 18.06 \times 10^{-24} = \frac{6.626 \times 10^{-34} \times 3 \times 10^{8}}{\lambda}$

$\lambda = \frac{6.626 \times 3 \times 10^{-26}}{18.06 \times 10^{-24}} = 1.1 \times 10^{-2} m$

(4) (a) $n = 3,\ l = 2,\ m_l = -2$

(b) $$P(r) = \int_{r=0}^{\infty} R^* R r^2 dr \int_0^{\pi}\int_0^{2\pi} Y^* Y \sin\theta \, d\theta \, d\phi$$

$$= \int_{r=0}^{\infty} R^* R r^2 dr \times 1$$

$$P(r)dr = R^* R r^2 dr$$

$$P(r) = R^* R r^2 = (R \times r)^2$$

$$= \left[\left(\frac{Z_f}{a_0}\right)^{3/2}\left(\frac{1}{2430}\right)^{1/2} \rho^2 \exp\left(-\frac{\rho}{2}\right)\right]^2 \times r^2$$

$$= \underbrace{\left(\frac{Z_f}{a_0}\right)^3 \times \frac{1}{2430} \times \left(\frac{2Z_f}{3a_0}\right)^4}_{A = \text{Const}} r^4 \exp\left(-\frac{2Z_f}{3a_0} r\right) \times r^2$$

$$= A r^6 \exp\left(-\frac{2Z_f}{3a_0} r\right)$$

$$\frac{dP(r)}{dr} = A \frac{d}{dr}\left[r^6 \exp\left(-\frac{2Z_f}{3a_0} r\right)\right]$$

$$= A \exp\left(-\frac{2Z_f}{3a_0} r\right)\left[6r^5 - \frac{2Z_f}{3a_0} r^6\right] = 0$$

$$\frac{2Z_f}{3a_0} r^6 = 6r^5 ; \quad \boxed{r = \frac{9a_0}{Z_f}}$$

$$\therefore \frac{2Z}{3a_0} r^6 = 6r^5 \; ; \quad r = \frac{9a_0}{Z}$$

(c) $$\langle r \rangle = \int_0^\infty R_{nl}^* \, r \, R_{nl} \, r^2 dr \int_0^\pi \int_0^{2\pi} Y_{2,-2} \, Y_{2,2} \sin\theta \, d\theta \, d\phi$$

$$= \int_0^\infty [R_{nl}]^2 r^3 dr$$

$$= \left(\frac{Z}{a_0}\right)^3 \cdot \frac{1}{2430} \left(\frac{2Z}{3a_0}\right)^4 \int_0^\infty r^7 \exp\left(\frac{-2Zr}{3a_0}\right) \cdot dr$$

$$= \left(\frac{Z}{a_0}\right)^7 \cdot \frac{1}{2430} \cdot \left(\frac{2}{3}\right)^4 \frac{7!}{\left(\frac{2Z}{3a_0}\right)^8}$$

$$= \left(\frac{Z}{a_0}\right)^7 \cdot \frac{1}{2430} \cdot \left(\frac{2}{3}\right)^4 \left(\frac{3}{2}\right)^8 \left(\frac{Z}{a_0}\right)^{-8} \cdot 7!$$

$$= \left(\frac{a_0}{Z}\right) \cdot \frac{1}{2430} \cdot \frac{3^4}{2^4} \cdot 7! = \frac{Z}{a_0} \cdot \frac{1}{30} \cdot \frac{7!}{2^4}$$

$$= \left(\frac{a_0}{Z}\right) \cdot \frac{1}{30} \cdot \frac{2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7}{2^4} = \frac{21}{2} \frac{a_0}{Z} = 10.5 \frac{a_0}{Z}$$

$$2Fe^{3+} + 3I^- \longrightarrow 2Fe^{2+} + I_3^{\ominus}$$

$$-\frac{d[Fe^{3+}]}{dt} = \frac{k[Fe^{3+}]^2 [I^-]^2}{[Fe^{3+}] + k'[Fe^{2+}]}$$

As the rate law consists of two terms in the denominator it will have atleast two steps, with the first step being reversible.

$$\rightleftharpoons \qquad \text{— (1)}$$

$$\longrightarrow \qquad \text{— (2)}$$

$[Fe^{2+}]$ comes in product. ~~$[Fe^{3+}]$ in~~ $[Fe^{3+}]$ behaves as reactant and an intermediate is formed. $[Fe^{2+}]$ must have formed in step (1) and $[Fe^{3+}]$ in step (2)

$$\rightleftharpoons Fe^{2+} + Int \qquad \text{— (1)}$$

$$Fe^{3+} + Int \longrightarrow \qquad \text{— (2)}$$

Considering (2) as slow step we can neglect $[Fe^{3+}]$.

$$\frac{k[Fe^{3+}]^2[I^-]^2}{k'[Fe^{2+}]}$$

Atom Fe = 2 - 1 = 1

I = 2

One Fe is already present in reactant so, intermediate is I_2^-

Balancing the ~~change~~, charge, $+6 - 2 - 2 = +2$

$$\rightleftharpoons Fe^{2+} + I_2^- \qquad \text{— (1)}$$

$$Fe^{3+} + I_2^- \longrightarrow$$

Balancing atoms ~~x~~ and charge,

$$Fe^{3+} + 2I^- \rightleftharpoons Fe^{2+} + I_2^- \qquad \text{— (1)}$$

$$Fe^{3+} + I_2^- \longrightarrow Fe^{2+} + I_2 \qquad \text{— (2)}$$

The mechanism requires one more step:

$$I_2 + I^- \longrightarrow I_3^- \quad \text{---} (3)$$

~~Combining~~ Combining (1), (2) and (3), we get,

$$Fe^{3+} + 2I^- \longrightarrow Fe^{2+} + I_2^- \quad \text{---} (1)$$

$$Fe^{3+} + I_2^- \longrightarrow Fe^{2+} + I_2$$

$$I_2 + I^- \longrightarrow I_3^-$$

$$2Fe^{3+} + 3I^- \longrightarrow 2Fe^{2+} + I_3^-$$

⑥

$$-\frac{d[A]}{dt} = k_1[A][B] - k_{-1}[I][D]$$

$$\frac{d[I]}{dt} = k_1[A][B] - k_{-1}[I][D] - k_2[I][B] + k_{-2}[C][D]$$
$$\simeq 0$$

$$[I] = \frac{k_1[A][B] + k_{-2}[C][D]}{k_{-1}[D] + k_2[B]}$$

$$-\frac{d[A]}{dt} = k_1[A][B] - k_{-1}\frac{k_1[A][B][D] + k_{-2}[C][D]^2}{k_{-1}[D] + k_2[B]}$$

$$= \frac{k_1k_{-1}[A]\cancel{[B]}[D] + k_1k_2[A][B]^2 - k_1k_{-1}\cancel{[A]}[B][D] + k_{-1}k_{-2}[C][D]^2}{k_{-1}[D] + k_2[B]}$$

$$= \frac{k_1k_2[A][B]^2 - k_{-1}k_{-2}[C][D]^2}{k_{-1}[D] + k_2[B]}$$

Q07

$$k_1 + k_2 = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{12.8} = 0.054 h^{-1}$$

$$\frac{k_1}{k_2} = \frac{62}{38}$$

$$k_2 = \frac{0.05415}{1 + \frac{62}{38}} = 0.02 h^{-1}$$

$$k_1 = \frac{0.05415}{1 + \frac{38}{62}} = 0.0336 h^{-1}$$

ACML100
Major, 2024
Marks: 40
(Prof. Shashank Deep, Prof. Nidhi Jain, Prof. Subrata Kundu)

(Important Note: There will be step wise marking. Make sure that every step is written in answer sheet. Skipping of any step will result into deduction of mark.

Questions 8-11 choose the correct answer (1.5 x 4 = 6)

(8) If Δ_0 is the octahedral splitting energy and P is the pairing energy, then crystal field stabilization energy (CFSE) of $[Co(H_2O)_6]Cl_2$ is

(a) $-0.8\Delta_0 + 2P$ (b) $-0.8\Delta_0 + 1P$ (c) $-0.8\Delta_0$ (d) $-1.8\Delta_0 + 1P$

(9) The complex $[Fe(Phen)_2(NCS)_2]$ (phen = neutral nitrogen based chelating ligand) shows spin cross-over behavior ($T_{1/2} = 175$ K). CFSE (excluding pairing energy) and μ_{eff} at 250 and 100 K respectively are

(a) $-0.4\ \Delta_0$, 4.90 BM and $-2.4\ \Delta_0$, 0.00 BM (b) $-2.4\ \Delta_0$, 2.90 BM and $-0.4\ \Delta_0$, 1.77 BM
(c) $-2.4\ \Delta_0$, 0.00 BM and $-0.4\ \Delta_0$, 4.90 BM (d) $-1.2\ \Delta_0$, 4.90 BM and $-2.4\Delta_0$, 0.00 BM

(10) Among the complexes (A) $[Mn(H_2O)_6]Cl_2$, (B) $K_4[Cr(CN)_6]$, and (C) $K_3[Co(CN)_6]$ Jahn-Teller distortion is expected in

(a) only in A (b) A, B, C (c) only in B (d) B and C

(11) Consider the following steps of a catalytic reaction and state the correct oxidation state of the metal complexes A, B and C respectively

$$\underset{(A)}{(PPh_3)_2Pd(CH_3)_2} \xrightarrow{CH_3I} (B) \xrightarrow{-CH_3CH_3} (C)$$

(a) 2, 0, 2 (b) 2, 4, 2 (c) 2, 4, 4 (d) 2, 3, 2

Questions 12-17: Comment the statements as true or false (6 x 1 = 6)

(12) $KMnO_4$ exhibits charge transfer transitions; therefore, it has lower intensity and a lower molar extinction coefficient. (F)

(13) Among the following metal complexes—$K_3[Mn(CN)_6]$ **(A)**, $K_3[Fe(CN)_6]$ **(B)**, and $K_3[Co(CN)_6]$ **(C)**—the central metal ion with the lowest ionic radius is found in complex **B**.
(F)

(14) According to VSEPR theory, the structure of XeF_4 is square planar, and the structure of XeF_2 is linear. (T)

(15) The total number of metal-metal bonds found in $Co_4(CO)_{12}$ is six (T)

(16) Mn_3O_4 has inverse spinal structure. (F)

(17) $Ni(CO)_4$, $[Co(CO)_4]^-$, and $[Fe(CO)_4]^{2-}$ follow the 18-electron rule; therefore, their carbonyl stretching frequencies (ν_{CO}) in the IR spectrum will be the same. (F)

Questions 18-21: Give descriptive answer (4 x 2 = 8)

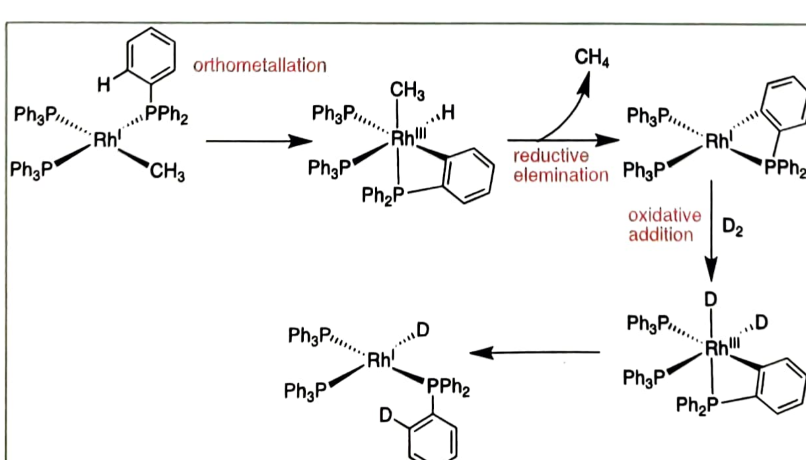
(18) Explain the Mond process used for the purification of nickel.

The Mond process purifies nickel by converting impure nickel into volatile nickel carbonyl, $Ni(CO)_4$, at around 50–60°C. The gas is then decomposed at higher temperatures (200–250°C) to yield pure nickel metal and release carbon monoxide for reuse.

Low temperature $Ni(CO)_4$ formation: 0.5 marks
Ni(CO)4 is Volatile: 0.5 marks
Decomposition of $Ni(CO)_4$ at high temperature: 0.5 marks
Reuse of carbon monoxide: 0.5 marks

If all the steps are described, a pictorial representation will also be awarded marks.

(19) The complex $(PPh_3)_3Rh(Me)$ in the presence of D_2 forms methane and $(PPh_3)_2Rh(D)(PPh_2C_6H_4D)$. Write the two important steps involve in this reaction.



(20) Among $[Fe(SCN)_3(H_2O)_3]$, $[NiCl_4]^{2-}$, and $[Mn(H_2O)_6]Cl_2$, which will exhibit low-intensity color and which will exhibit high-intensity color, and why?

Low

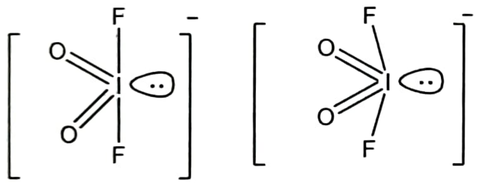
$[Mn(H_2O)_6]Cl_2$ Low intensity color (0.5 marks)
In $[Mn(H_2O)_6]Cl_2$, the electronic transition is Laporte forbidden and spin forbidden (0.5 marks)
$[Fe(SCN)_3(H_2O)_3]$: high-intensity color (0.5 marks)
In complex $[Fe(SCN)_3(H_2O)_3]$, charge transfer occurs from ligand to metal and charge transfer transition are Laporte allowed and spin allowed. (0.5 marks)

High

(21) (a) Write the geometry and structure of $[ClO_2]^-$ and comment on the deviation of the O-Cl-O bond angle from the ideal geometry. (b) Write the structure of $[IO_2F_2]^-$.

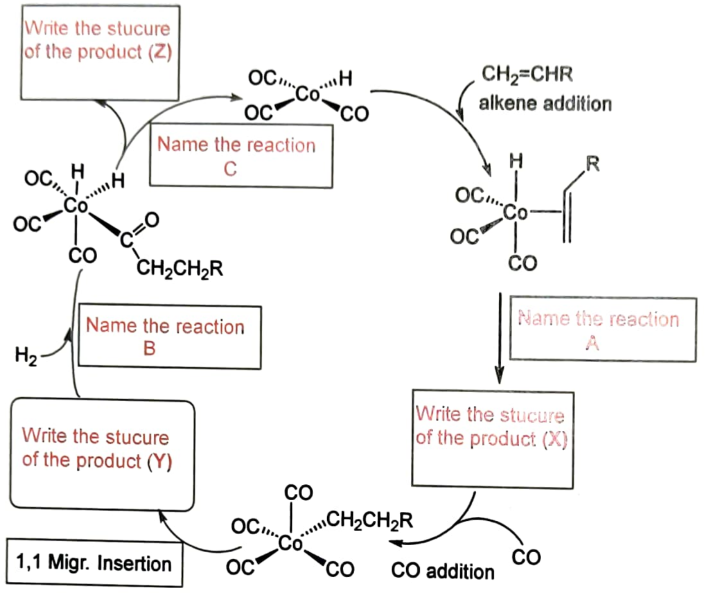
(a) $[ClO_2]^-$: Tetrahedral geometry and V-shape (0.25 + 0.25 marks); O-Cl-O bond angle should be less than ideal tetrahedral bond angle that is 109.5° (0.5 marks)

(b)

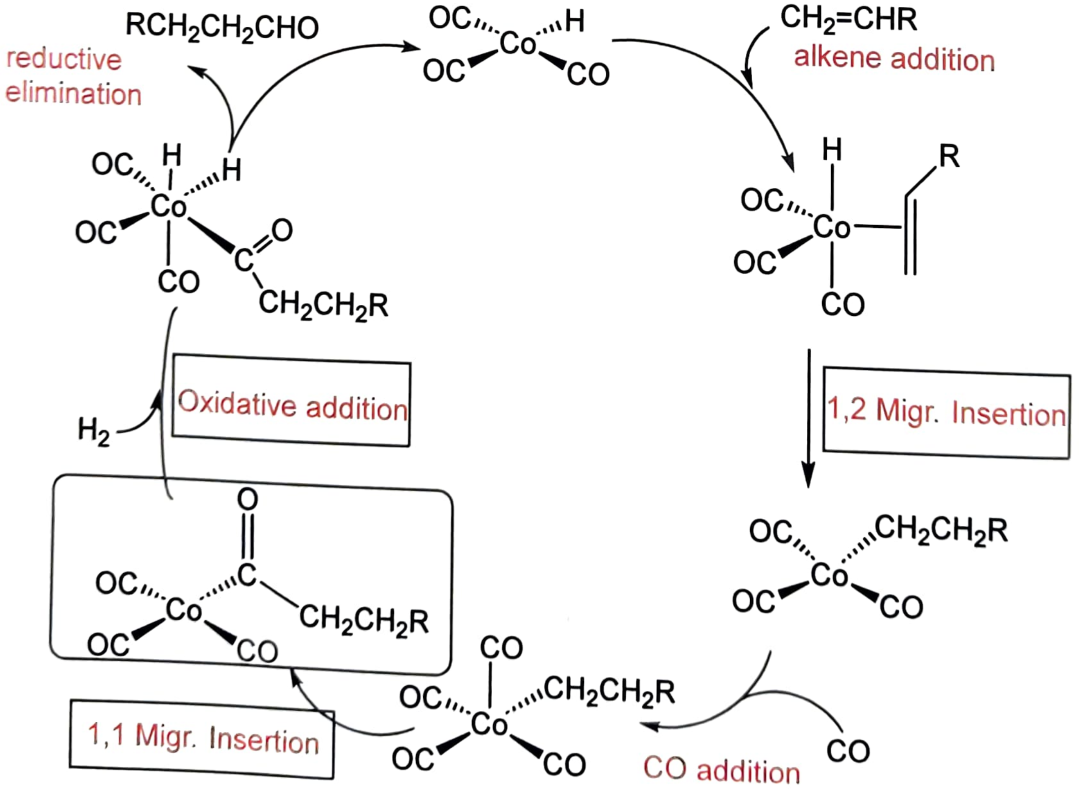


1 marks if deviation from the ideal geometry is visible or explaned
0.5 marks if only ideal geometry is descibed or drawn

Question 22: Find out the missing steps [3]
A catalytic cycle for a hydroformylation reaction is shown below. In this cycle, the names of the reactions (A, B and C) and the structures of the products (**X**, **Y** and **Z**) in three steps are missing. Provide the names of the missing reactions (A, B and C) and the structures of the products (**X**, **Y** and **Z**).



There are 6 unknowns (A, B, C and **X**, **Y**, **Z**): all have 0.5 marks each.



....End..